Lawal Olarotimi BADRU

lobadru@crimson.ua.edu awwal-badru.github.io +1.205.886.0907
LinkedIn Google Scholar ORCID

Summary

Computer Science educator with formal pedagogical training (B.Ed. in Computer Science) and university-level teaching experience. Strong applied computing background with expertise in Software Engineering, Computer Science, Applied AI, and full-stack development, supported by intelligent systems research and real-world system design. Experienced in teaching diverse undergraduate learners, designing structured instructional materials, and mentoring students in applied projects. Prepared to teach core CS courses including programming, software engineering, data structures, algorithms, cybersecurity, artificial intelligence/machine learning, and human-computer interaction. Committed to applied learning, inclusive pedagogy, and the teacher-scholar model.

Education

2023 – present **PhD Candidate (ABD) in Computer Science, University of Alabama**
Advisor: Prof. Jeffrey Carver
Dissertation: Learning Beyond the Lecture: Rethinking Software Engineering Education with Supplementary Materials in the Generative AI Era
Expected Completion: May 2027

2025 **MS in Computer Science, University of Alabama**
Comprehensive Exam: Effective Methods for Teaching Software Engineering
Relevant Coursework: Network Security, Software Security, Software Supply Chain Security, Cryptography, IoT Security, Computer Communication and Networks, Empirical Software Engineering, Algorithm

2023 **MSc by Research in Computing Science, University of the South Pacific**
Supervisor: Dr. Vani Vasudevan
Thesis: Intelligent Education Management Information System (IEMIS) for Fiji

2008 **BEd in Computer Science, University of Ibadan**
Minor: Mathematics
First Class (Honours); Best Graduating Student, Computer Science Department

University Teaching Experience

2016 – 2019 **Assistant Lecturer**, D.S. Adegbenro ICT Polytechnic, Ogun State, Nigeria

- Independently taught undergraduate courses in computing, mathematics, and information systems.
- Designed lectures, tutorials, laboratory exercises, and assessments aligned with curriculum standards.
- Used worked examples, guided practice, and structured progression to support diverse learners.
- Provided academic mentoring and supervised undergraduate projects.
- Courses taught: Advanced Calculus, Assembly Language, Object-Oriented Programming, Functions and Geometry, Mathematics Methods I & II, Management Information Systems.

University Teaching Experience (continued)

2014 – 2016 **Education Officer**, Staff Development Center, Bureau of Establishment & Training, Abeokuta, Ogun State, Nigeria

- Delivered digital literacy training for adult learners and administrative staff.
- Developed instructional materials for diverse learner backgrounds.

Teaching Strengths and Pedagogical Training

- Formal pedagogical training through a B.Ed. in Computer Science
- Experience teaching first-generation and underprepared learners
- Skilled in designing structured lessons, worked examples, and formative assessments
- Strong ability to anticipate learning gaps and adjust instruction accordingly
- Experience developing course materials and contributing to curriculum sequencing
- Strong academic advising and mentoring record (100+ students)
- Clear, intentional communication and inclusive instructional practices
- Experienced in hands-on, project-based instruction using real-world systems and industry-relevant tools

Research Overview (Teaching-Connected)

My research focuses on Applied AI, Software Engineering, Human-Computer Interaction, and Intelligent Systems, with an emphasis on adaptive learning platforms, GenAI-based evaluators, and applied computing systems. I study how students interact with computing materials and how intelligent tools can support learning, decision-making, and engagement. My work contributes to intelligent systems design, sequential decision-making, applied machine learning, and computing education research.

Publications

Journal Articles

1. **L. Badru** and J. Carver, "Personalizing supplementary materials in software engineering: Impact of preferences, learning styles, and gender," *ACM Transactions on Computing Education*, vol. 26, no. 4, 2026. DOI: 10.1145/3806053
2. **L. Badru**, V. Vasudevan, M. Khan, and G. Lingam, "Mern stack web-based education management information systems for pacific island countries," *SN Computer Science*, vol. 4, no. 1, p. 70, 2022. DOI: 10.1007/s42979-022-01437-0
3. **L. Badru**, V. Vasudevan, M. Khan, and G. Lingam, "State of education management information system (emis) in pacific melanesia small island developing states (sids)," *Directions: Journal of Educational Studies*, vol. 35, no. 2, pp. 107–118, 2021.

Conference Proceedings

1. **L. Badru** and J. C. Carver, "Alp: A lightweight adaptive learning platform that helps students find the right explanation," in *Proceedings of the 58th ACM Technical Symposium on Computer Science Education (SIGCSE TS 2027)*, Accepted demonstration paper; proceedings forthcoming, 2027.

2. **L. Badru**, M. An, J. Stamper, and J. Carver, "Optimizing learning: A comparative study of adaptive experiments and randomization in a software engineering course," in *Proceedings of the Sixth Annual Workshop on A/B Testing and Platform-Enabled Learning Engineering (PELE), Learning@Scale 2025*, Palermo, Italy, 2025.

Preprints

1. **L. Badru** and J. Carver, *How undergraduate software engineering students navigate learning materials across contexts*, Under Review ACM Transactions on Computing Education (TOCE), Jun. 2026.

Work-in-progress

1. **L. Badru** and J. Carver, *Personalized support: Evaluating a digital platform for supplementary materials in software engineering education*, Manuscript in preparation for submission to ACM Transactions on Computing Education (TOCE), 2026.
2. **L. Badru** and J. C. Carver, *What students use and what works: A systematic review and automated pedagogical evaluation of supplementary materials in software engineering*, Manuscript in preparation for submission to Computers & Education, 2026.

Employment History

2023 – present — **Research Assistant**, Department of Computer Science, University of Alabama

- Conducted empirical studies involving undergraduate software engineering students.
- Designed adaptive learning experiments across multiple universities.
- Analyzed student learning behaviors and developed supplementary materials.
- Contributed to ACM publications and data analysis workflows

2016 – 2022 — **Senior Education Officer / Acting Head, State EMIS**, Ministry of Education, Science and Technology, Abeokuta, Nigeria

- Oversaw statewide EMIS operations supporting 1,500+ schools and 500,000+ students.
- Developed analytics tools improving reporting accuracy and reducing processing time.
- Trained 1,000+ personnel as UNICEF Master Trainer.
- Contributed to long-term educational planning and policy simulation initiatives.

Employment History (continued)

2016 – 2019 **Assistant Lecturer (Secondment)**, D.S. Adegbenro ICT Polytechnic, Nigeria

- Served as primary instructor with responsibility for curriculum sequencing, assessment design, and grade moderation.
- Developed comprehensive tutorials, lecture notes, and practical laboratory exercises.
- Facilitated mastery of complex topics using structured lessons and worked examples.
- Supervised undergraduate projects and mentored students in computing, mathematics, and information systems.

2014 – 2016 **Education Officer (Secondment)**, Staff Development Center, Bureau of Establishment & Training, Ogun State, Nigeria

- Delivered training in computer applications and digital literacy for administrative staff.
- Developed instructional materials for diverse learner backgrounds.

2010 – 2014 **Programmer Analyst**, Teaching Service Commission, Abeokuta, Nigeria

- Supported educational data systems and ICT capacity-building initiatives.

Workshop and Training

May, 2026 **Workshop Facilitator**, Tai Solarin Federal University of Education, Ijagun, Ogun State, Nigeria

- Led two-day virtual workshop on Agentic Coding for advanced CS students.

Skills

Teaching & Pedagogy	Lecture design, assessment development, worked examples, Project-based learning and inclusive instruction
Computing Education Research	Experimental Experimental design, mixed methods, learning analytics, instructional material evaluation
Programming & Systems	Python, JavaScript, Dart, PHP; Flutter, React, Node.js; UNIX/Linux fundamentals
Databases	MySQL, PostgreSQL, MongoDB
Tools & Methodologies	Git/GitHub, Firebase CLI & Emulators, Google Maps Platform, ESLint
AI & Data:	Machine learning fundamentals, personalization models, sequential decision-making, A/B testing, Alteryx, Tableau, Pandas, NumPy, SciPy
Human-Centered Computing	Interface design, user studies, interpretability considerations in adaptive systems

Miscellaneous Experience

Scholarly Contribution

2026 **Reviewer**, *Special Interest Group on Computer Science Education (SIGCSE)*. Review manuscripts in Computing Education Research (CER).

2022 **Reviewer**, *Education and Information Technologies* (Springer Nature). Reviewed manuscripts in educational technology and learning analytics.

Applied Computing & Industry Projects

2026–present **Real-Time Geolocation Emergency Alert System**. *Technologies* Flutter, Dart, Firebase (Firestore, Cloud Functions, FCM), Riverpod, Google Maps API, Node.js.

- Engineered a cross-platform real-time emergency dispatch application enabling SOS alerts and responder coordination.
- Implemented geospatial querying and proximity matching for dispatching nearest responders.
- Built serverless backend workflows using Firebase Cloud Functions and Firestore listeners.
- Provides experience with real-time, cloud-based Information Systems integrating geospatial data and intelligent dispatch workflows.

2026 **Financial Management Information System (FMIS)**. *Technologies* MongoDB, React.js, Express.js, Node.js.

- Contributed to requirements analysis, system design, database modeling, and iterative development.
- Strengthens practical experience in Information Systems development, secure data handling, and operational decision support.

Awards and Scholarships

2021 **Research Grant**, Queen Elizabeth Commonwealth Scholarship, United Kingdom (3,000 USD)

2020 – 2021 **Commonwealth Scholarship**, United Kingdom (value over 50,000 USD), University of the South Pacific, Fiji

2019 **Dean's Award for Academic Excellence & Global Leader Scholarship**, University of Bath, United Kingdom (value £10,000)

2008 **Best Graduating Student**, Department of Computer Science, Tai Solarin University of Education, Nigeria (value 500 USD)

Certifications

2021 **Predictive Analytics for Business**, Udacity (Alteryx & Tableau)

Microsoft Azure Fundamentals, Microsoft

Ideas for a Better World: Leading Change Through Policymaking, British Council

Mentorship

2010 – 2026 Advised and mentored over 100 senior secondary and university students, supporting academic and professional development.

2026 **Judge**, Alabama Robotics Contest, University of Alabama. Evaluated robot navigation accuracy, task completion, and engineering design.

Miscellaneous Experience (continued)

Future plan

- Mentor undergraduates in adaptive systems, contextual bandits, Explainable AI, and human-centered computing, providing well-scoped research projects aligned with my scholarly agenda.

Professional Memberships

- **Association for Computing Machinery (ACM)**
- **Institute of Electrical and Electronic Engineers (IEEE)**
- **Special Interest Group on Computer Science Education (SIGCSE)**

References

Available on Request